

Optimization using calculus

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- Objective 2: Demonstrate the level of rigor needed at our level. Everything must be substantiated.
- Key lesson: You can often learn a lot, and catch possible errors, by taking the time to *inspect the problem* and *reflect on your solution*.

Univariate problems

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- Sanity check – reflect on your solution. Hint: Looking at the max problem, the constraint is $x \geq 0$...

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 - A condition is either satisfied or it is not satisfied.

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- Exercise in the handout.

- A reminder of second-order conditions in two dimensions.

Bivariate problems

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- A note on expectations when graphing functions.

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- We need

$$f_{xx} \leq 0, f_{yy} \leq 0, \text{ and } f_{xx}f_{yy} - (f_{xy})^2 \geq 0.$$

It cannot be overstated how important the second part is, yet it is surprisingly often overlooked. This is one of the few things you need to memorize.

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 - The function is concave only if $a \in [-2, 2]$.

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$f(0, 0) = 0$ the function value at the candidate solution

$f(z, z) = (a - 2)z^2$ when x and y take the same value (z)

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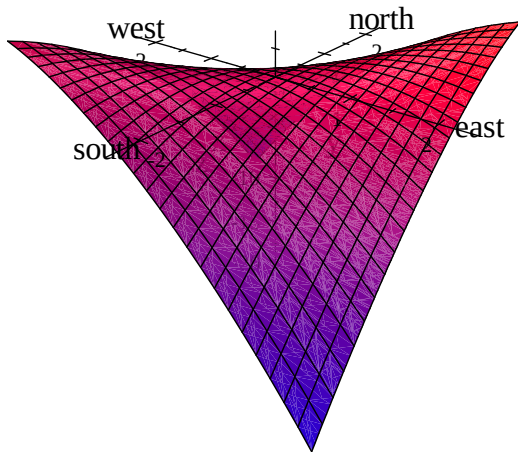
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- For any $z \neq 0$: $f(z, z) > 0$ when $a > 2$ and $f(z, -z) > 0$ when $a < -2$!

Concavity and SOC in two dimensions

- For $a = 3$, the function increases as we move to the north-east (or south west).



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- Similar for an indifference curve: You can sometimes solve $f(x, y) = k$ for y . This gives you y as a function of x (and k), which you can now sketch.

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- The fine print: Reexamining the necessary conditions.

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- Caveats:
 - 1 Don't lose sight of the economics. Following a recipe can obscure intuition.
 - 2 Do not forget to think logically about the problem at hand. The Lagrangian method can be slow. *Inspect the problem and look for shortcuts.*

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- We now formulate the Lagrangian,

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 - We cannot sign the multiplier to the equality constraint, $g(\mathbf{x}) = b$. It is unclear if a change in b changes the feasible set in a beneficial way.
- We could add the nonnegativity constraints directly to the Lagrangian. However, it is more common to adjust the optimality conditions.

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- Subject to some fine print, the **necessary** Kuhn-Tucker conditions: If $\mathbf{x}$ is feasible and a maximum, then there are multipliers such that

$$\frac{\partial \mathcal{L}}{\partial x_i} \leq 0, \text{ with } \frac{\partial \mathcal{L}}{\partial x_i} = 0 \text{ if } x_i > 0, i = 1, 2, \dots, n$$
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 - If $\lambda_h > 0$ then $h(\mathbf{x}) = c$.
 - If $h(\mathbf{x}) < c$ then $\lambda_h = 0$.
 - Don't overlook: Both $\lambda_h = 0$ and $h(\mathbf{x}) = c$ *at the same time* is also possible.

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- Since there are often several candidate solutions, the typical approach can involve a lot of work.
 - *A shortcut would be nice.*

A quick review

- Since the Kuhn-Tucker conditions are necessary conditions:
 - Identify all $(\mathbf{x}, \lambda_g, \lambda_h)$ that satisfy the Kuhn-Tucker conditions plus the constraints in the maximization problem.
 - There are often several candidate solutions.
 - The true maximum can be found by checking which has the highest $f(\mathbf{x})$ value.
- Since there are often several candidate solutions, the typical approach can involve a lot of work.
 - *A shortcut would be nice.*
 - Thus, *inspecting the problem* may prove useful.

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- *Inspect the problem:* If x_1 and/or x_2 is zero, then utility is zero. Any interior bundle gives higher utility.
 - It is sufficient to look for an interior solution, $x_1, x_2 > 0$. There is a *unique* such solution.

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 - The utility function is symmetric in x_1 and x_2 . Hence the solution is to buy the cheapest good only.

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 - Any number of inequality constraints, $h_k(\mathbf{x}) \leq c_k$, are allowed. Nonnegativity constraints can be captured in this way.
- **Theorem:** Assume there are no equality constraints. Assume that the objective function $f(\mathbf{x})$ is quasiconcave and that all $h_k(\mathbf{x})$ are quasiconvex. Then, if $\mathbf{x}^*$ satisfies the Kuhn-Tucker conditions and $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then $\mathbf{x}^*$ solves the problem.

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 - An implication is that the constraint qualification is **not** an issue in the standard utility maximization problem.

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 - But if the m constraints are linearly dependent, then we lose degrees of freedom.
 - Not enough information/freedom to solve Kuhn-Tucker.

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- Conclusion: Kuhn-Tucker does not find the correct solution.
 - At $\mathbf{x} = (1, 1)$, $\nabla h(\mathbf{x}) = (0, 0)^T$. This is a linearly dependent vector, so the constraint qualification does not hold.

Some exercises

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- A trickier utility function is

$$u(x_1, x_2) = (2x_1 + x_2)^{\frac{1}{2}} x_1.$$

Solution in the handout.